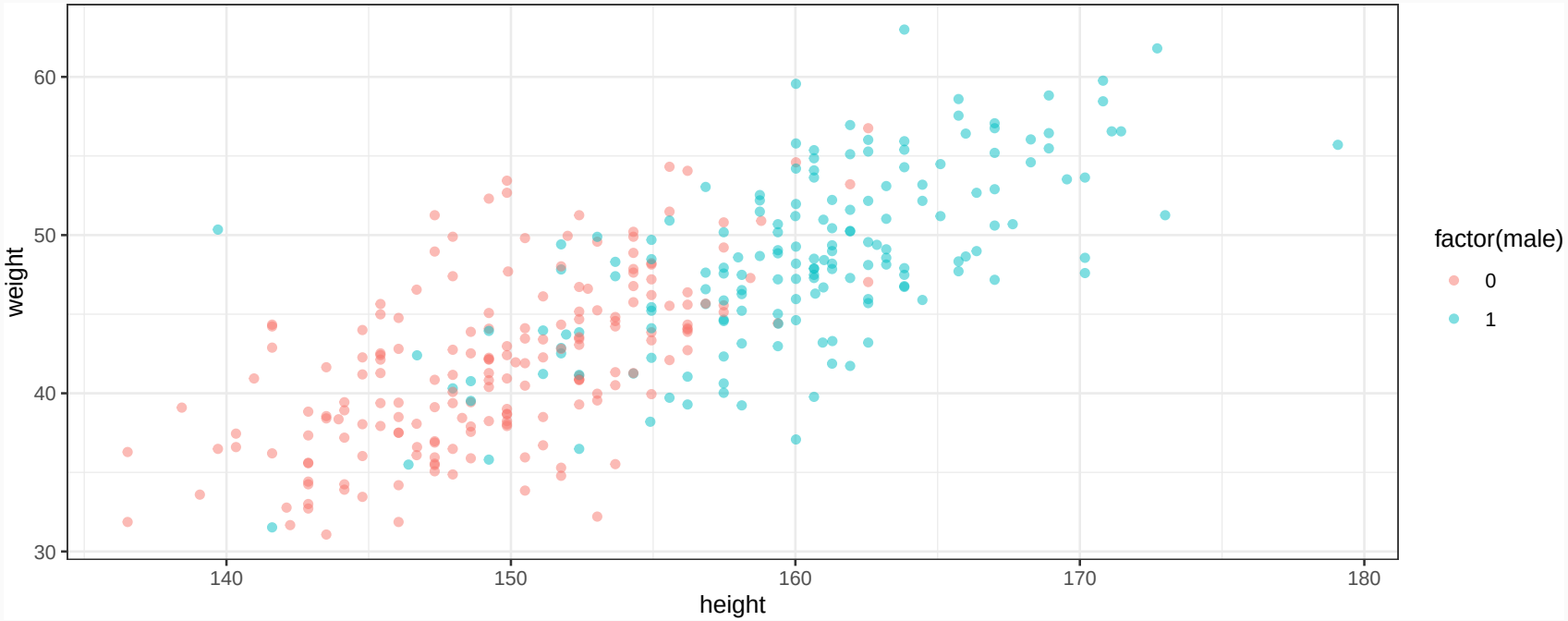
Structural causal models and multiple regression

Frank Edwards

Introduction to causal inference with graphs

Causality isn't obvious in data



$$H \rightarrow W$$

$$W \rightarrow H$$

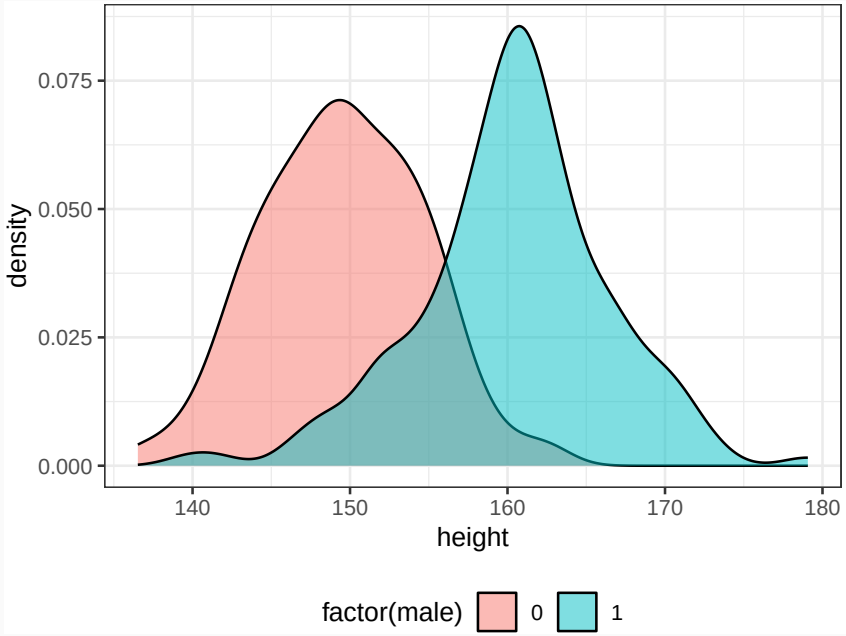
Causality isn't obvious in data

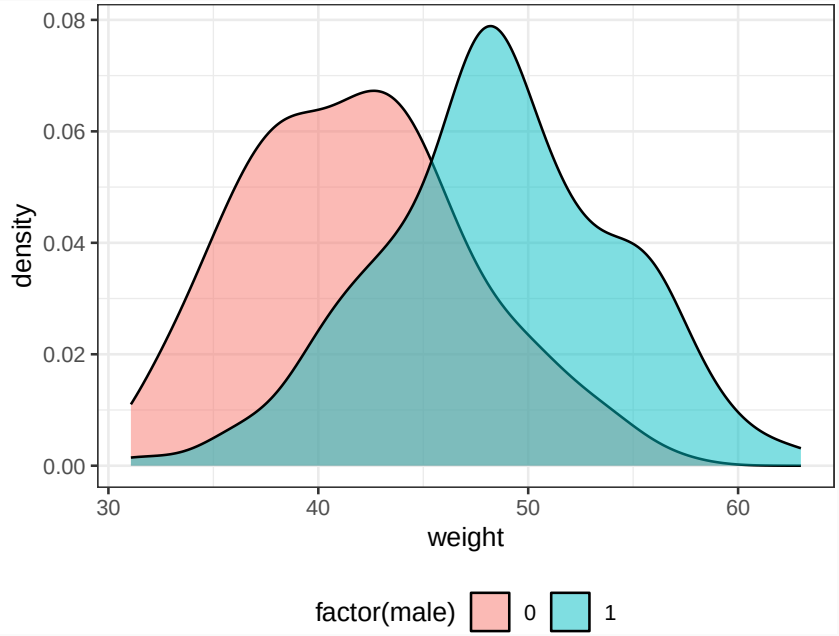
$$S \to W$$

$$W \to S$$

$$S \to H$$

$$H \to S$$





Describing our estimands

Estimands define the theoretical quantity we are interested in learning about. For this analysis, let's pick three:

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- The average total causal effect of sex on weight

Describing our estimands

Estimands define the theoretical quantity we are interested in learning about. For this analysis, let's pick three:

- The average causal effect of height on weight
- The average total causal effect of sex on weight
- The average direct causal effect of sex on weight

Drawing our theory

Height influences (*listens to*) weight. We draw an *edge* to connect the *nodes* H and W

Drawing our theory

Height influences (*listens to*) weight. We draw an *edge* to connect the *nodes* H and W

```
dagW<-dagitty("dag{
  H -> W
}")

ggdag(dagW, layout = "linear") + theme_dag()
```



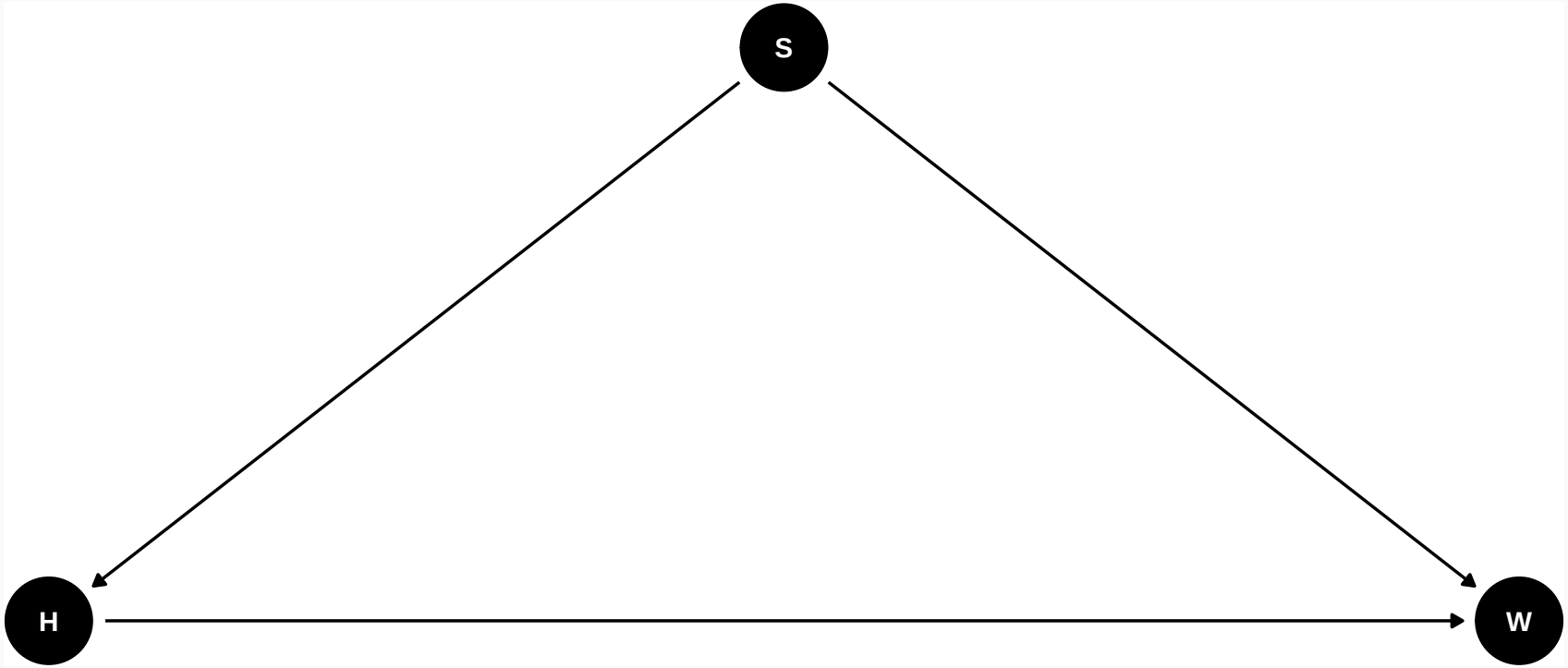
Drawing our theory

Sex influences (*listens to*) height and weight

Drawing our theory

Sex influences (*listens to*) height and weight

We draw edges to connect S to H and W



Directed acyclic graphs

- Reflect our scientific knowledge

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- Is a *graph* consisting of nodes and edges

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Directed acyclic graphs

- Reflect our scientific knowledge
- Is a *graph* consisting of nodes and edges
- Edges have a direction (arrow)
- Does not have any cycles (no closed loops)
- Nodes that are connected are *associated* (but functional form is not implied)

Family metaphor

We sometimes use a family tree metaphor to describe DAGs

```
dag1<-dagitty("dag{
  x -> y
  y -> z
}")
ggdag(dag1, layout = "linear") + theme_dag()
```



Family metaphor

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```
dag1<-dagitty("dag{
  x -> y
  y -> z
}")
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```



- z is a *descendant* of x and y

Family metaphor

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```
dag1<-dagitty("dag{
  x -> y
  y -> z
}")
ggdag(dag1, layout = "linear") + theme_dag()
```



- z is a *descendant* of x and y
- y and x are *ancestors* of z

Family metaphor

We sometimes use a family tree metaphor to describe DAGs

```
dag1<-dagitty("dag{
  x -> y
  y -> z
}")
ggdag(dag1, layout = "linear") + theme_dag()
```



- z is a *descendant* of x and y
- y and x are *ancestors* of z
- y is a *parent* of z, and z is a *child* of x

Family metaphor

```
ggdag_ancestors(dag1, "z", layout = "linear") + theme_dag()
```



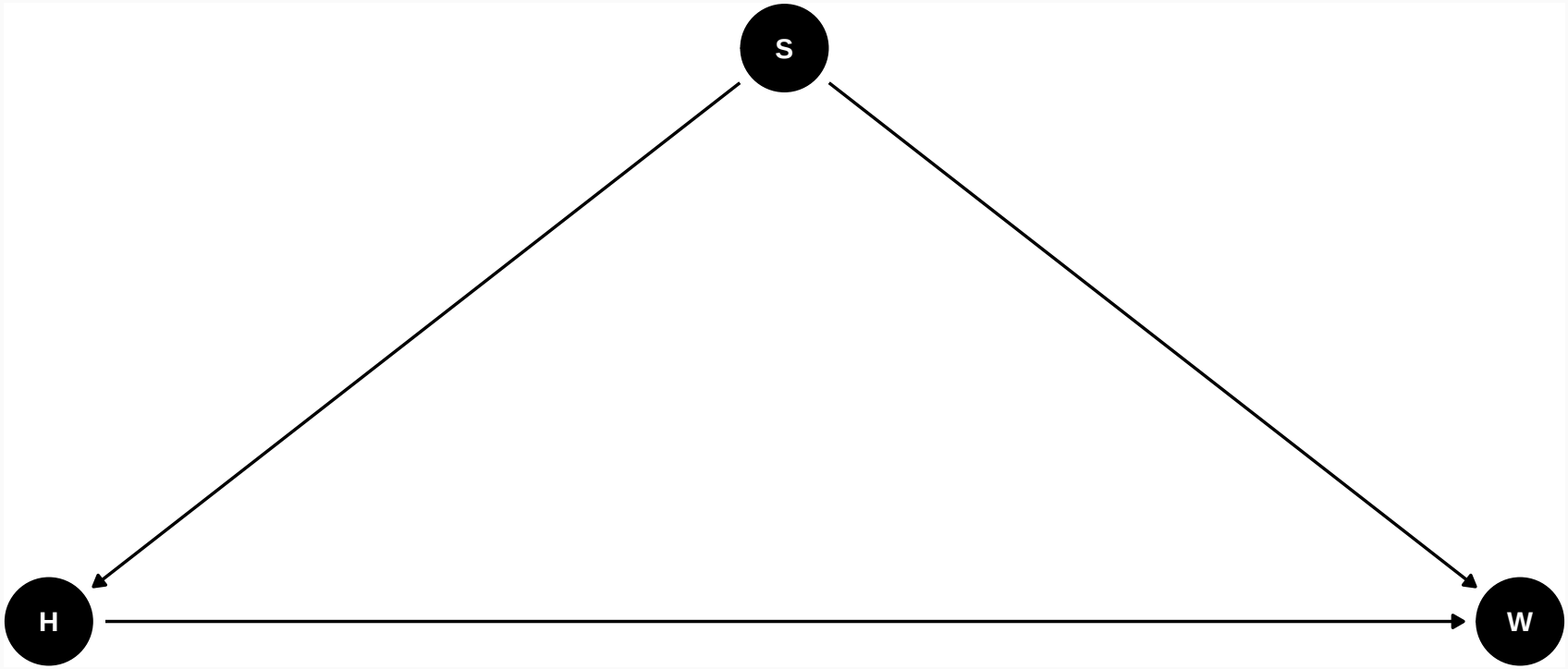
Family metaphor

```
ggdag_parents(dag1, "z", layout = "linear") + theme_dag()
```



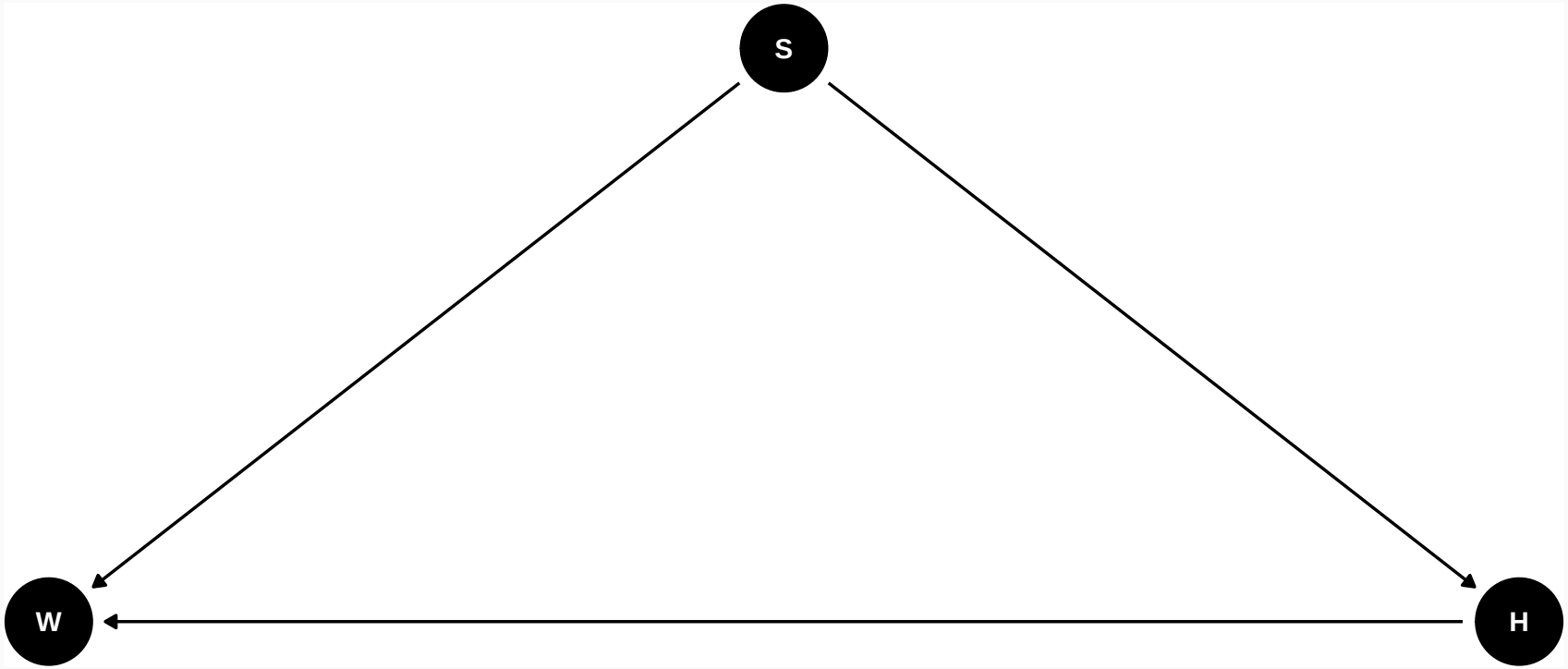
What it means (mathematically)

1. $H = f(S)$
2. $W = g(S, W)$



Asking a causal question with a DAG

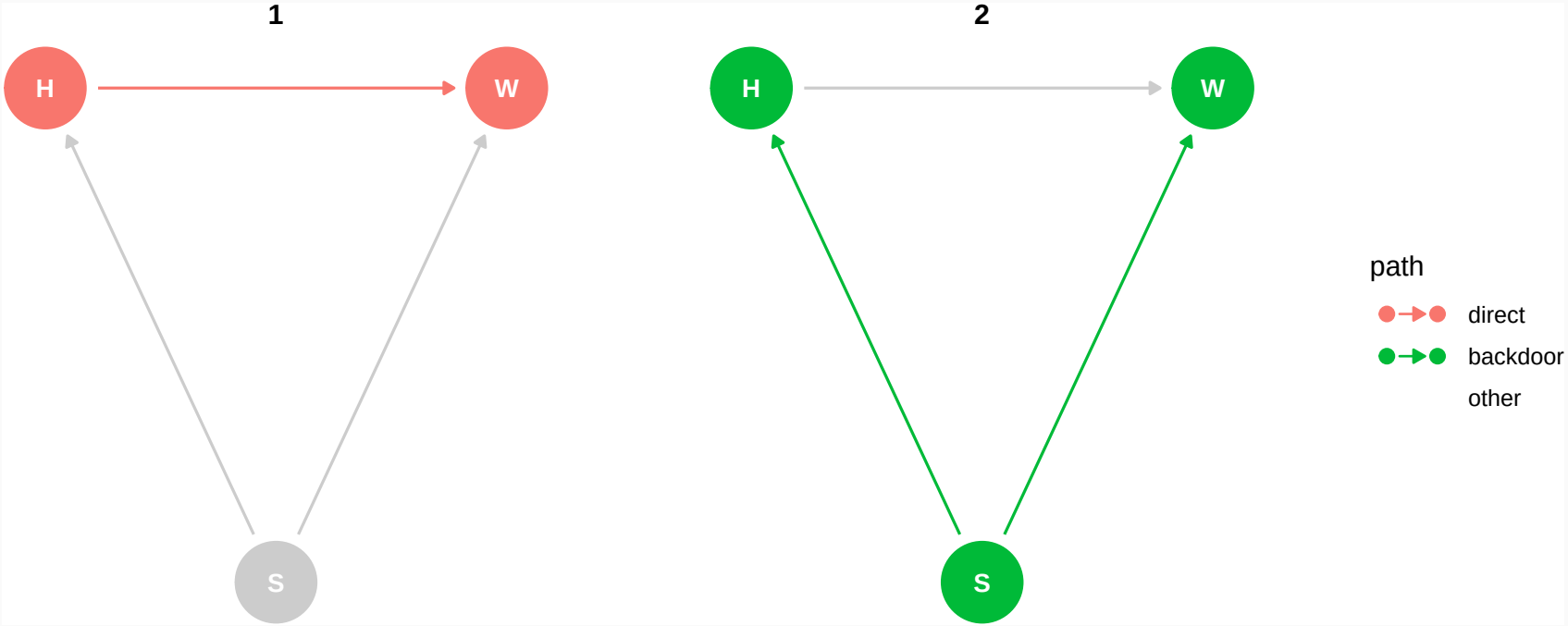
What is the causal effect H on W?



The backdoor criterion

Information flows between connected nodes. To identify an effect, we need to close all *backdoor paths* between the outcome and the exposure.

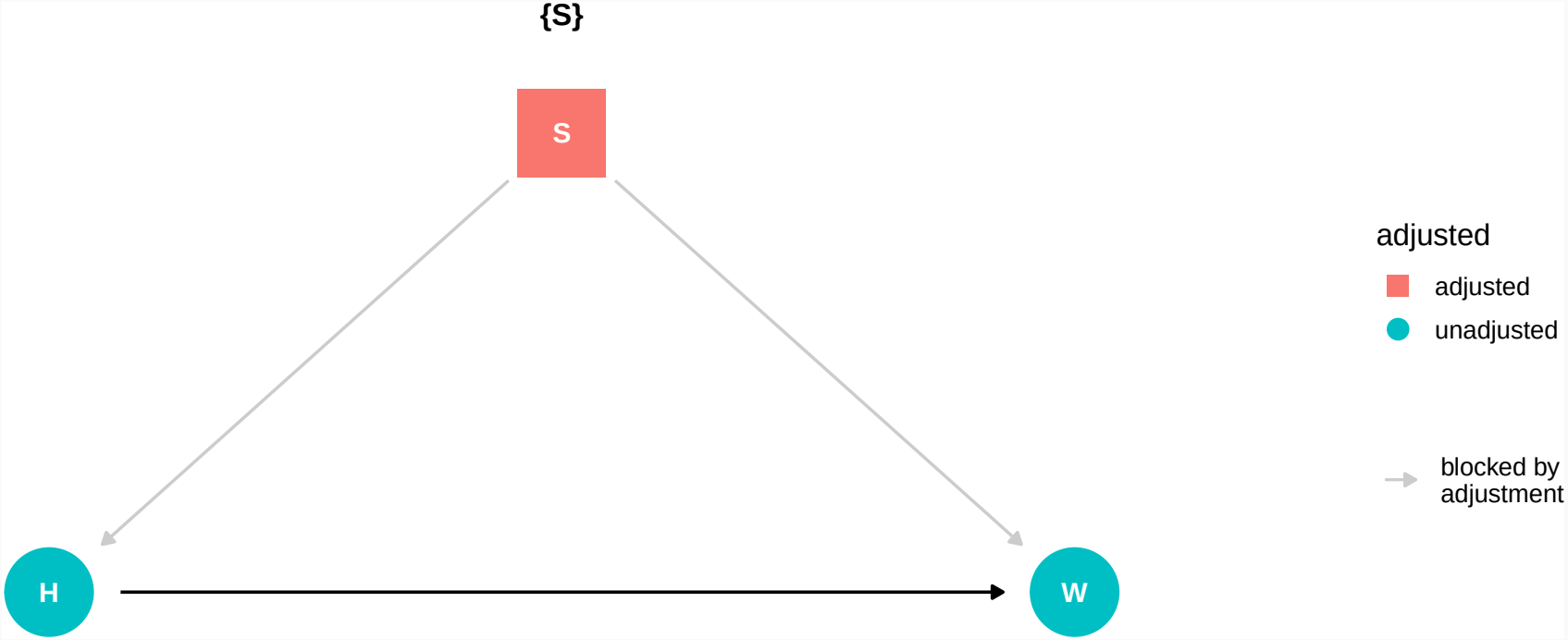
```
ggdag_paths(dagW) + theme_dag()
```



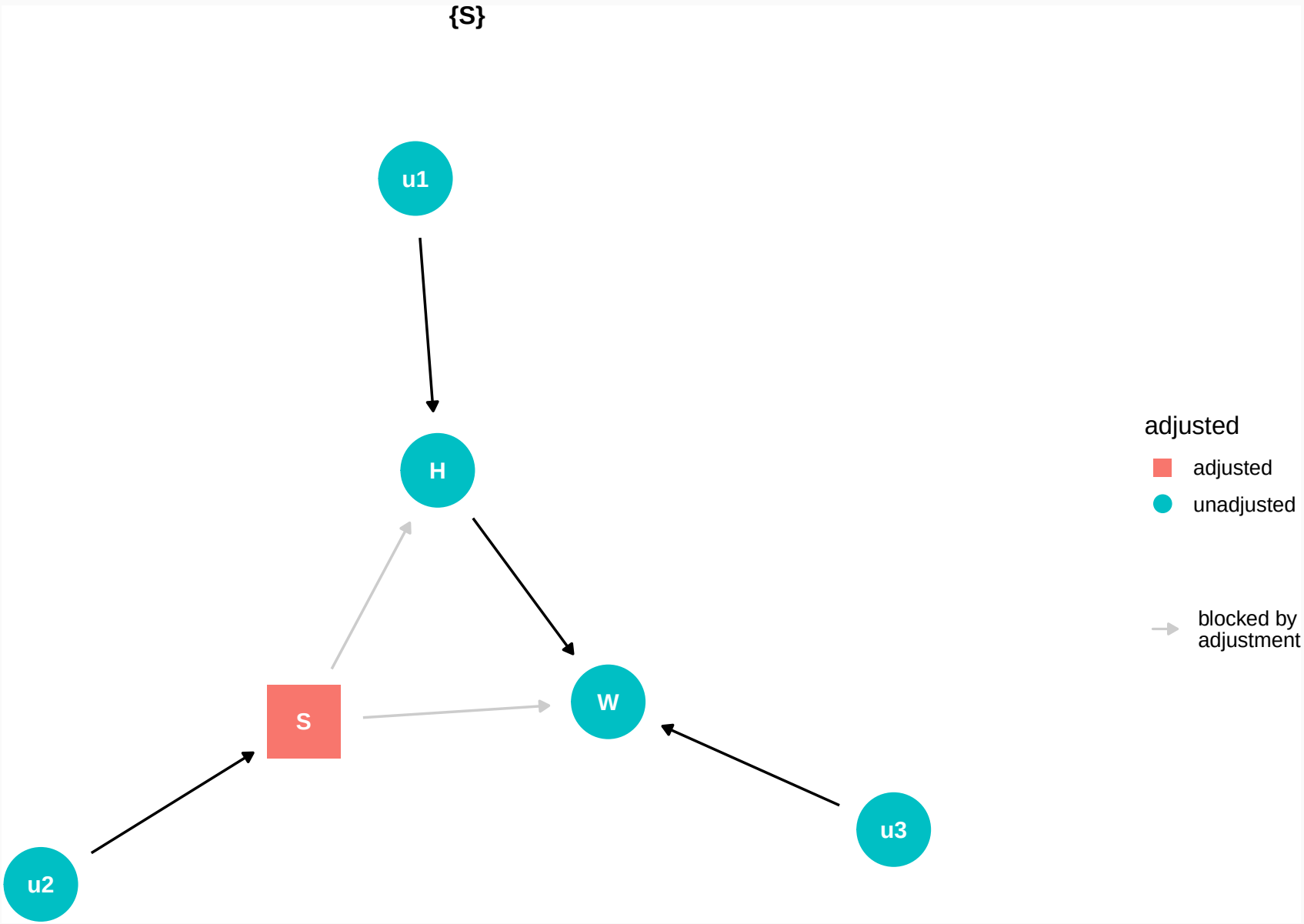
Closing backdoor paths

We close backdoor paths by *controlling* for nodes along a backdoor path

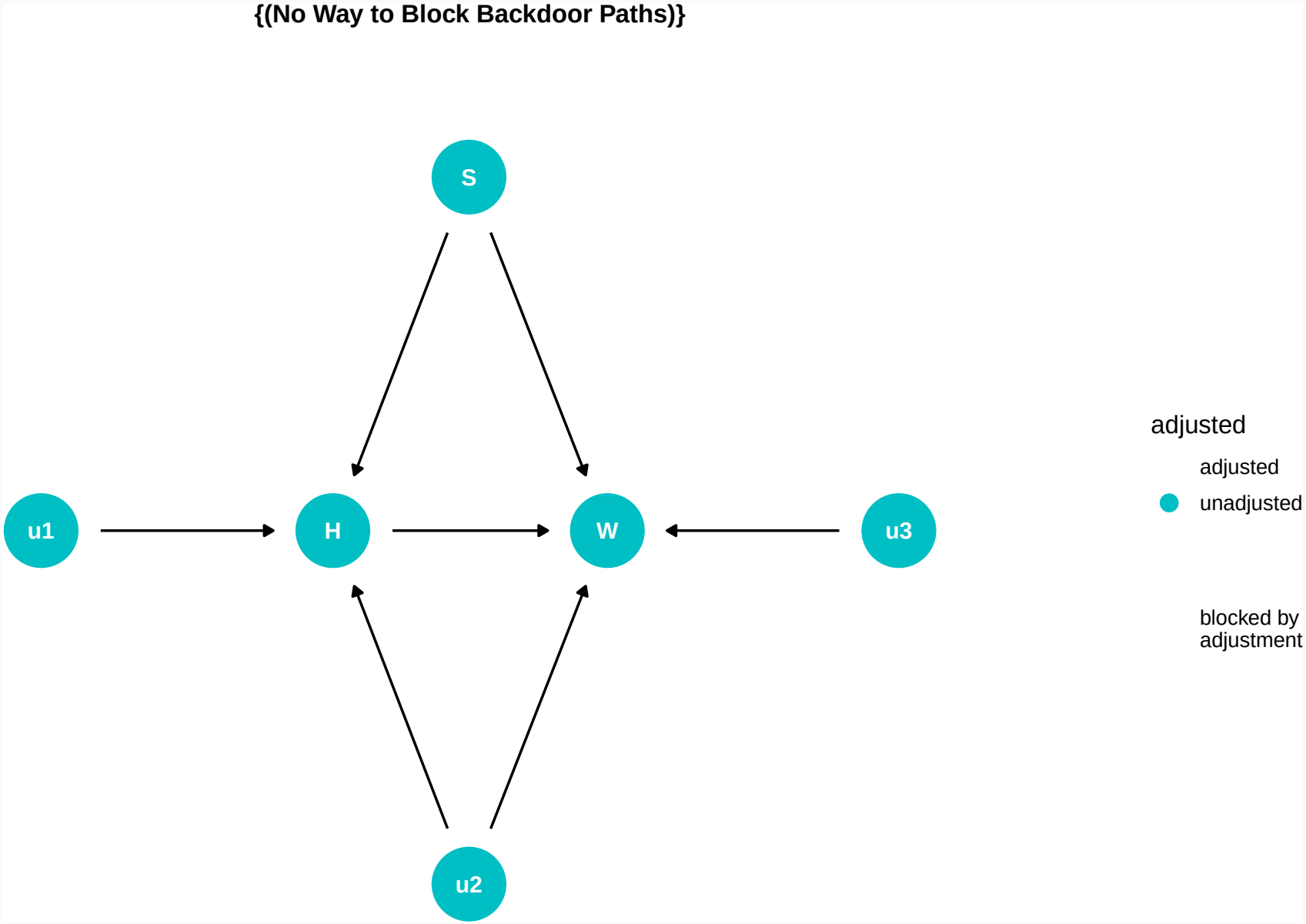
```
ggdag_adjustment_set(dagW) + theme_dag()
```



Unobserved influences (ignorable)



Unobserved influences (not ignorable)



Closing backdoor paths with regression

DAGs help us identify which variables to control. Simply close all backdoor paths through *conditioning* (controlling) to estimate $p(W|H, S = s)$

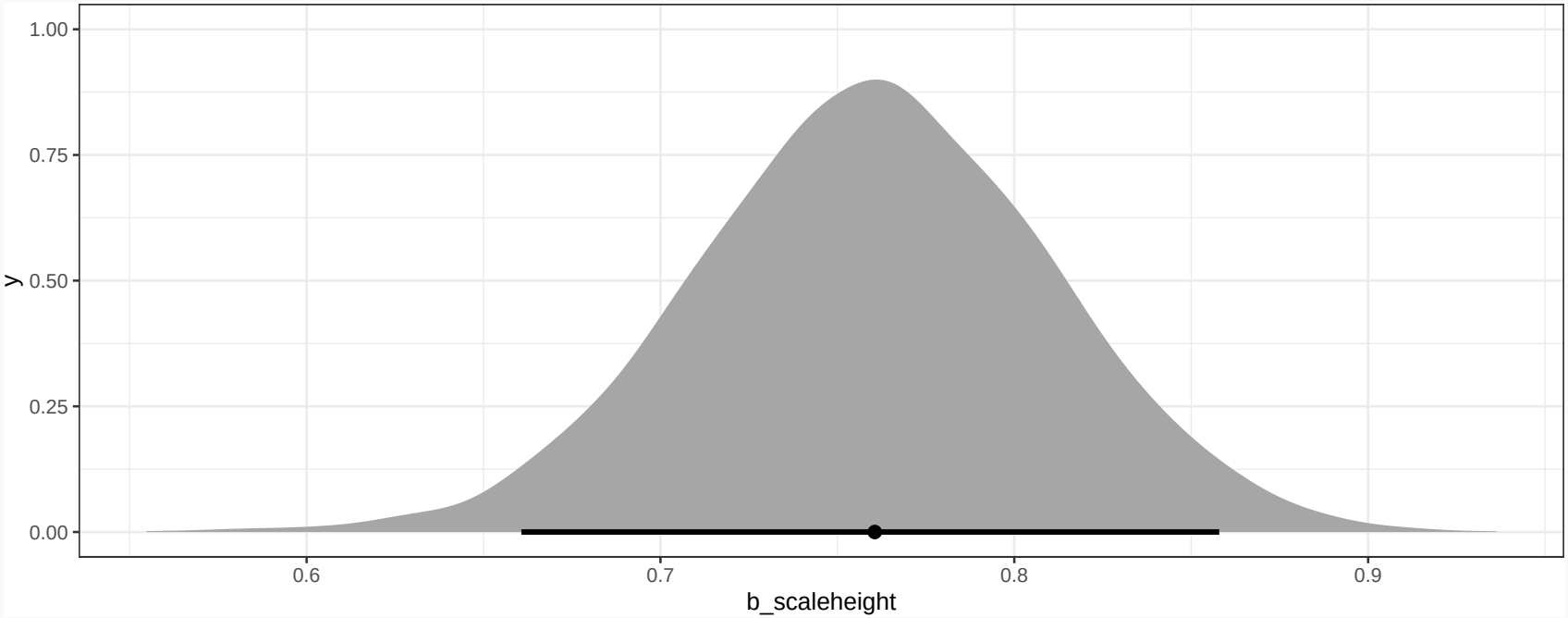
```
# identify minimal set of controls
adjustmentSets(dagW)
```

```
# set priors
priors <- c(set_prior("normal(0,1)", class = "b"),
            set_prior("exponential(1)", class = "sigma"))
# fit model
m_h <- brm(scale(weight) ~ -1 + # remove intercept
             factor(male) +
             scale(height),
           data = d)
```

Posterior inference of an effect

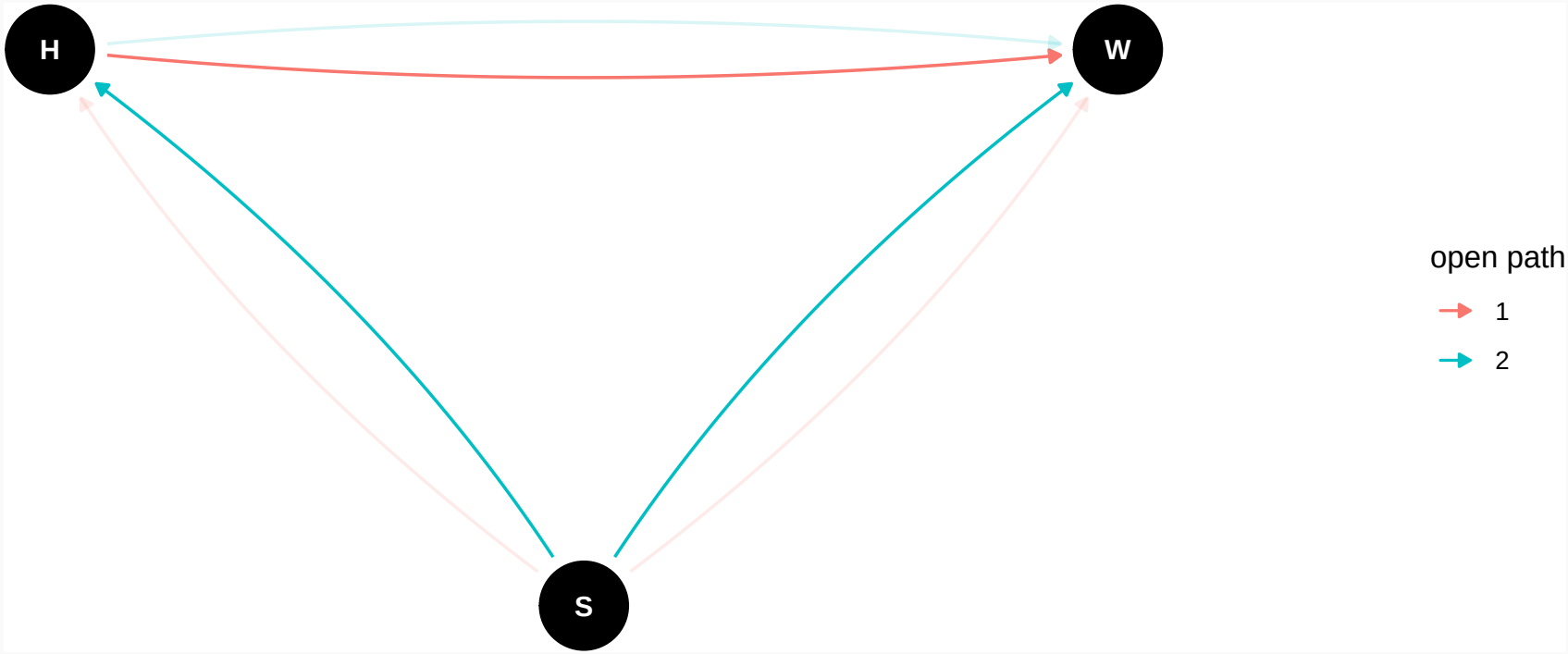
The mean posterior difference in average weight for a one standard deviation increase in height is easy to obtain, it is simply β.

```
post <- as_draws_df(m_h)
ggplot(post, aes(x = b_scaleheight)) +
  stat_slabinterval(.width = 0.95)
```



Another estimand

What is the causal effect of sex on weight?



The sum of *direct* and *indirect* effects is the *total effect*

The total effect

```
adjustmentSets(dagW_s)
```

```
{}
```

No adjustment! Just estimate $p(W|S)$ directly with regression

```
m_s <- brm(scale(weight) ~ -1 + # remove intercept
             factor(male),
           prior = priors,
           data = d)
```

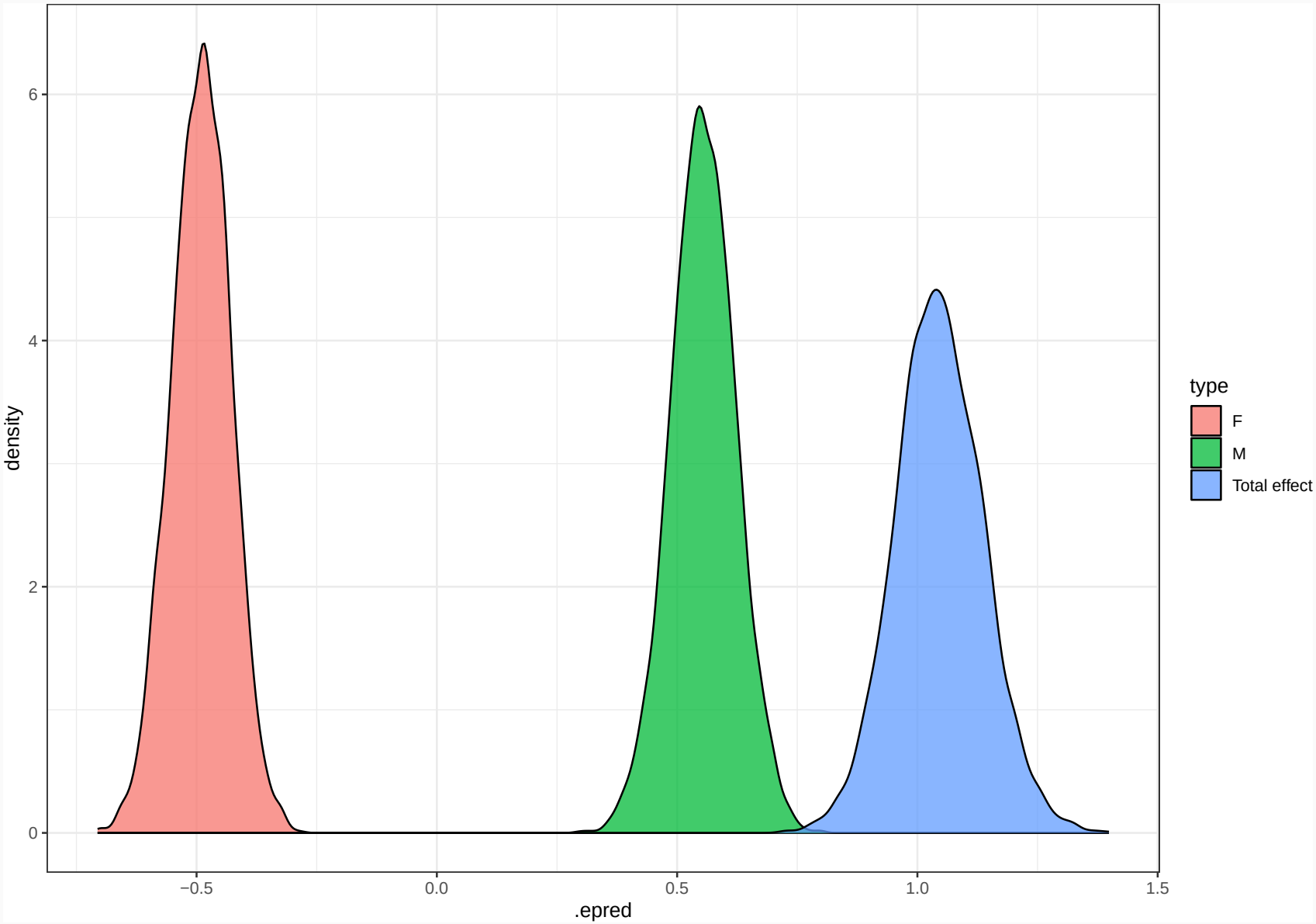
Posterior

We use the posterior distribution to compute a difference;

$$E[W|S = male] - E[W|S = female]$$

```
post <- data.frame(male = c(1,0)) |> add_epred_draws(m_s)
post_m <- post |> filter(male==1) |> mutate(type = "M")
post_f <- post |> filter(male==0) |> mutate(type = "F")
diff <- post_m |>
  mutate(.epred = .epred - post_f$.epred) |>
  mutate(type = "Total effect")
# plot
pdat <- post_f |>
  bind_rows(post_m) |>
  bind_rows(diff)
```

Posterior density for parameters and causal effect



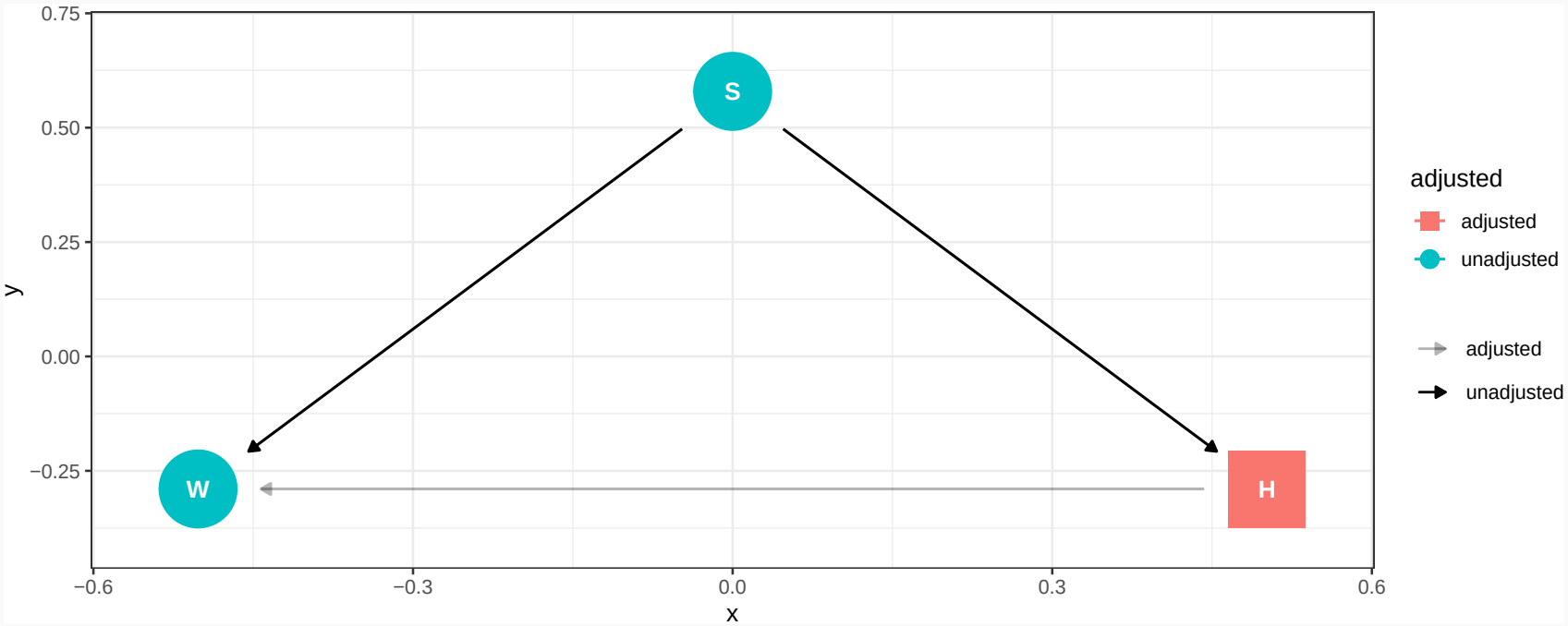
Kinds of questions we can ask using a DAG: Direct vs total effects

- What is the direct effect of a hypothetical manipulation of S on W, holding H constant $p(W|S, H = h)$

```
adjustmentSets(dagW_s, effect = "direct")
```

```
{ H }
```

```
ggdag_adjust(dagW_s, var="H")
```



Fitting the model

```
m_s_direct <- brm(scale(weight) ~ - 1 + # remove the intercept
                    factor(male) +
                    scale(height), # control for height
            prior = priors,
            data = d)
```

Regression as a decomposition of effect components

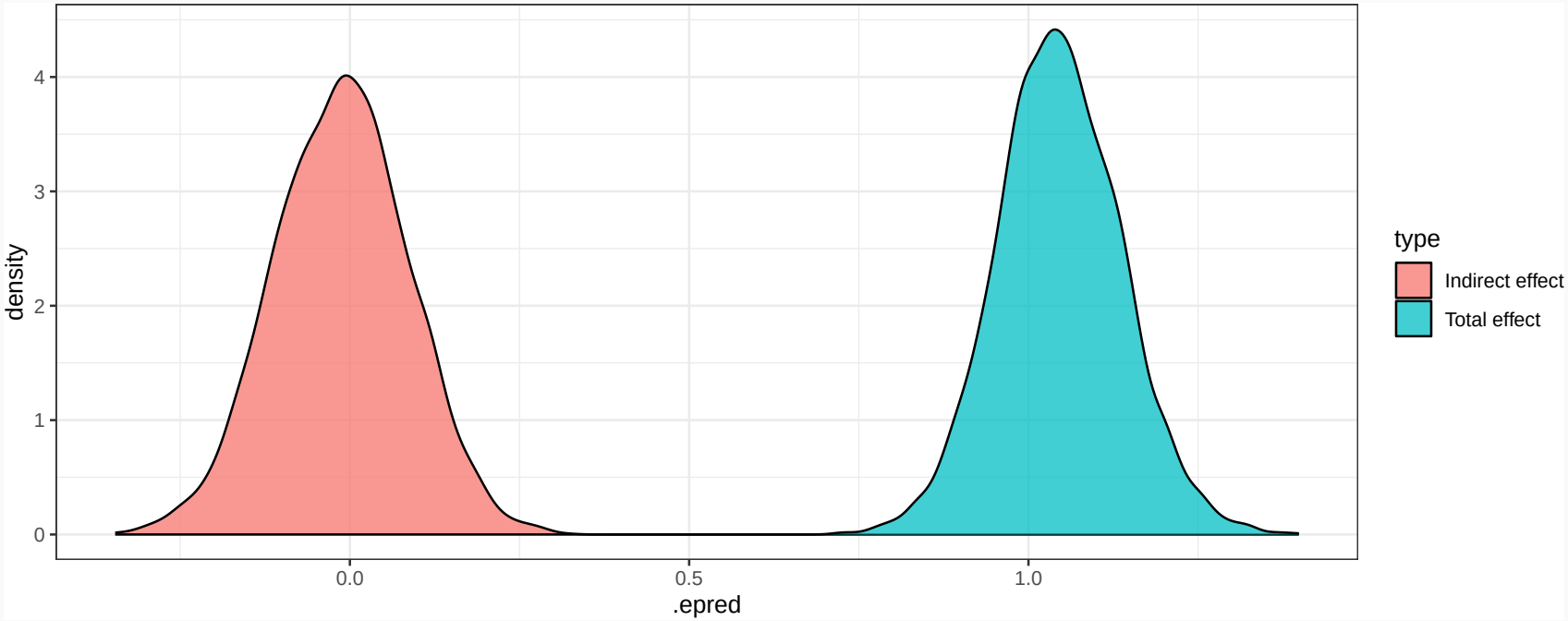
Compute contrast again, then diff out from the total effect estimate (diff)

```
# compute contrast
post <- data.frame(male = c(1,0), height = mean(d$height)) |>
  add_epred_draws(m_s_direct)
post_m <- post |> filter(male==1) |> mutate(type = "M")
post_f <- post |> filter(male==0) |> mutate(type = "F")
diff2 <- post_m |>
  mutate(.epred = .epred - post_f$.epred) |>
  mutate(type = "Indirect effect")
# what share of the total effect of sex is a 'direct' effect?
prop_direct <- diff2$.epred/diff$.epred
# posterior median and 80% interval
quantile(prop_direct, c(0.1, 0.5, 0.9))
```

```
        10%         50%         90%
-0.13346277 -0.01100691  0.10867280
```

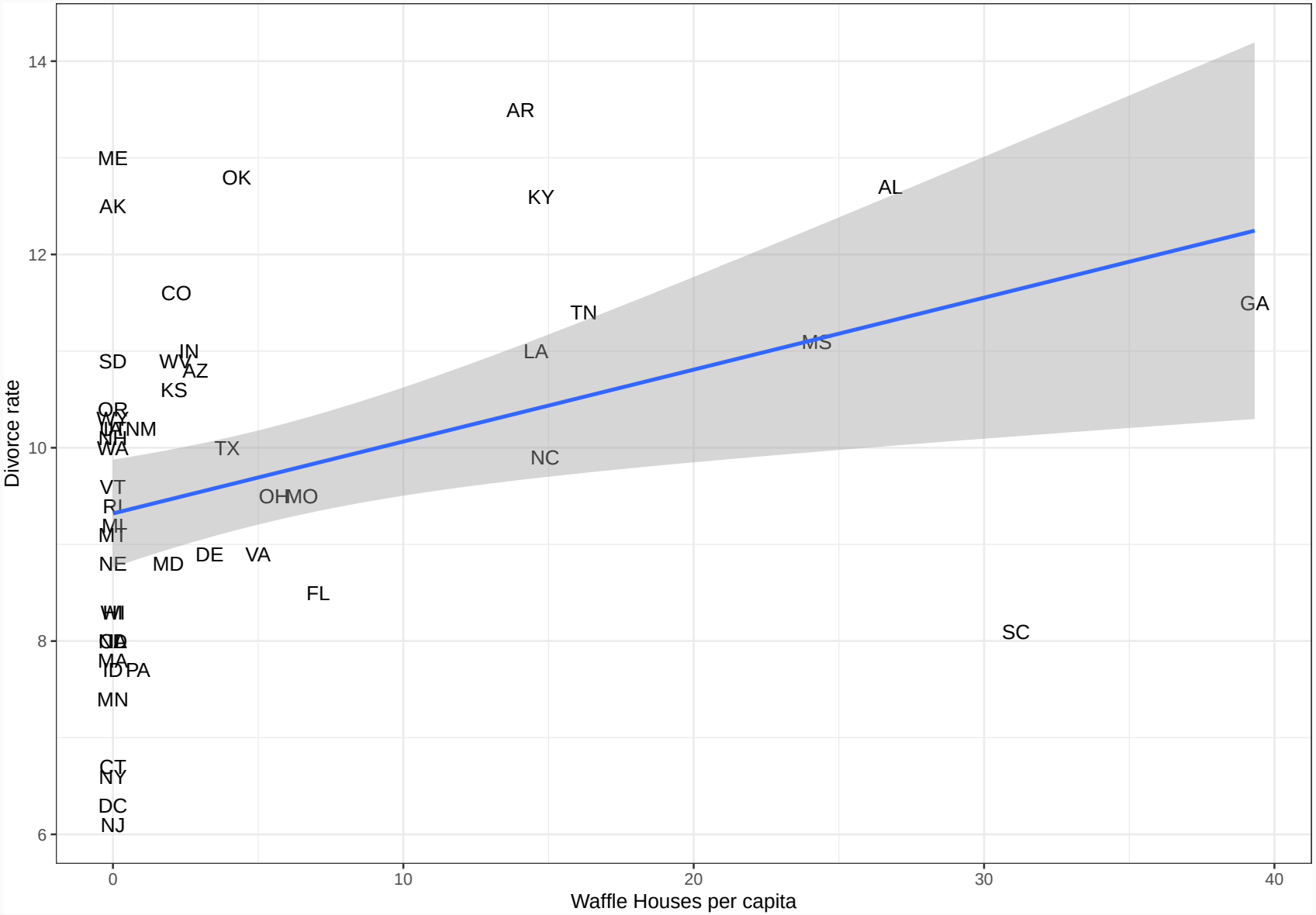
Visualize the posterior direct and total effect estimates (sex->height)

```
diff |>
  bind_rows(diff2) |>
  ggplot(aes(x = .epred, fill = type)) +
  geom_density(alpha = 0.75)
```



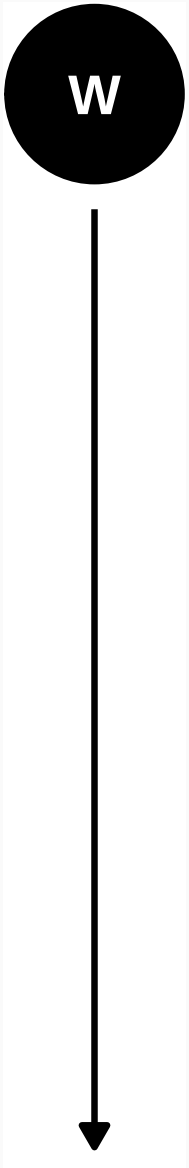
Multiple regression

Waffle house and divorce rates



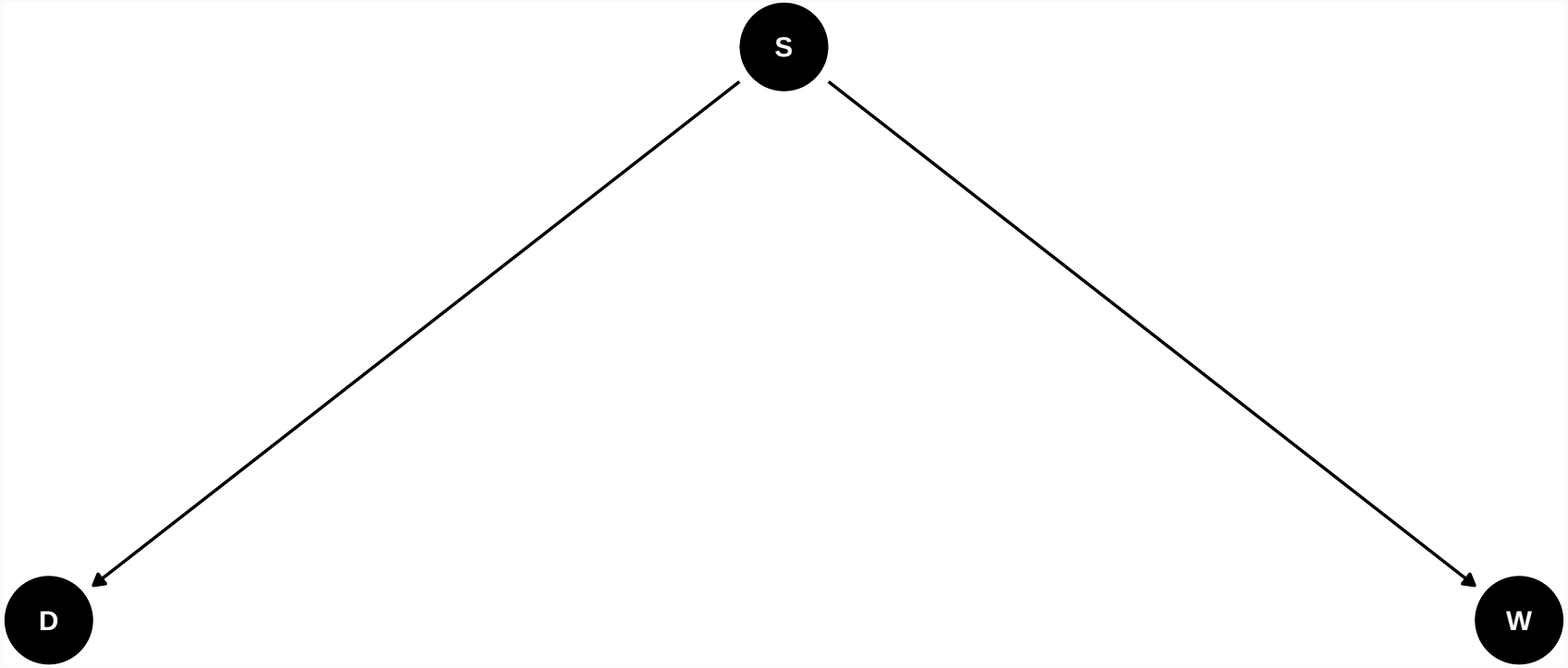
One potential causal theory

```
dagWaffle <- dagify(D ~ W, outcome = "D")
ggdag(dagWaffle) + theme_dag()
```



A confounding fork

```
dagWaffle <- dagify(D ~ S,
                    W ~ S, outcome = "S")
ggdag(dagWaffle) + theme_dag()
```



Two theories for the causes of divorce rates

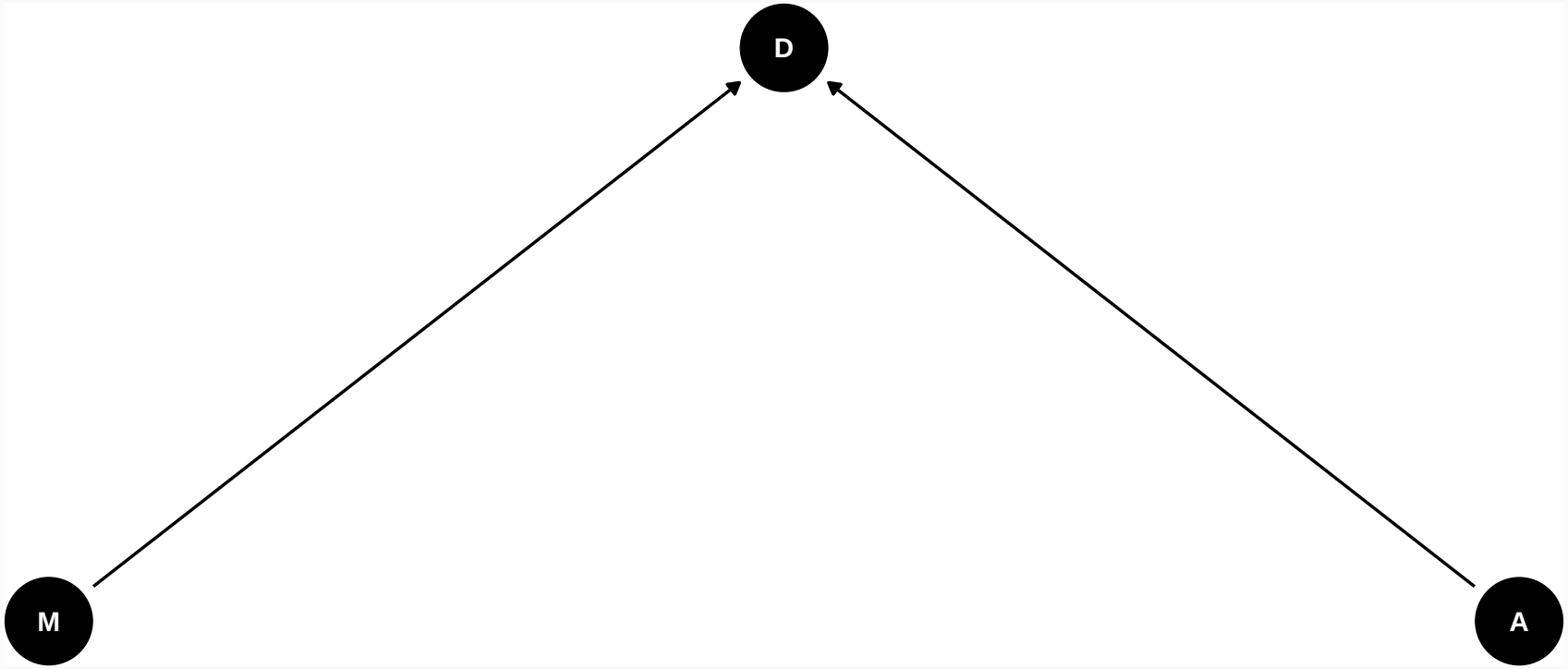
M: Marriage rate, D: Divorce rate, A: Median age at marriage

```
dagMarriage<-dagify(D~M, outcome = "D")
p1 <- ggdag(dagMarriage) + theme_dag()
dagAge <- dagify(D~A, outcome = "D")
p2 <- ggdag(dagAge) + theme_dag()
p1+p2
```



An equivalent statement

```
dagM <- dagify(D~M+A, outcome = "D")
ggdag(dagM) + theme_dag()
```



Propose a model for divorce rates with age at first marriage as a predictor

$$D_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_A A_i$$

$$\alpha \sim \text{Normal}(0, 0.2)$$

$$\beta_A \sim \text{Normal}(0, 0.5)$$

$$\sigma \sim \text{Exponential}(1)$$

Load and scale variables

Recall that scaling variables to $\bar{x} = 0, sd = 1$ makes defining priors and fitting complex models *much* easier.

```
WaffleDivorce<- WaffleDivorce  |>
  mutate(A = scale(MedianAgeMarriage),
         D = scale(Divorce),
         M = scale(Marriage))
```

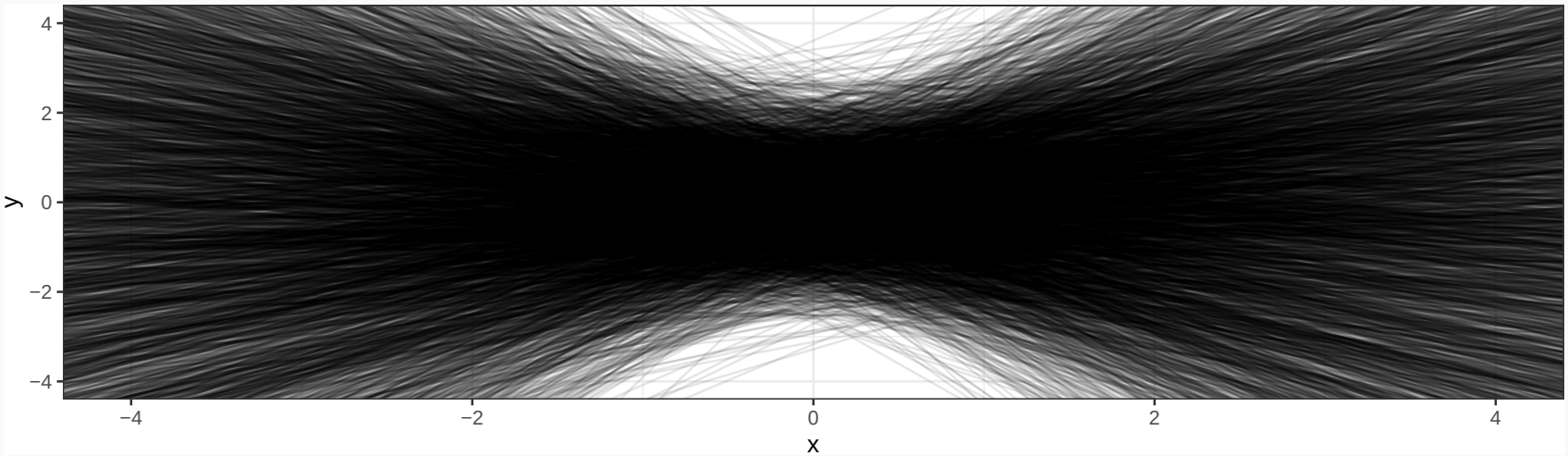
Fit the model(s)

```
priors <- c(
  set_prior("normal(0, 1)", class = "Intercept"),
  set_prior("normal(0, 1)", class = "b"),
  set_prior("exponential(1)", class = "sigma")
)
# add sample_prior = T to include prior draws for us to peek at
m_age <- brm(D ~ A,
             prior = priors,
             data = WaffleDivorce,
             sample_prior = T,
             silent = 2)

m_marriage <- brm(D ~ M,
                  prior = priors,
                  data = WaffleDivorce,
                  sample_prior = T,
                  silent = 2)
```

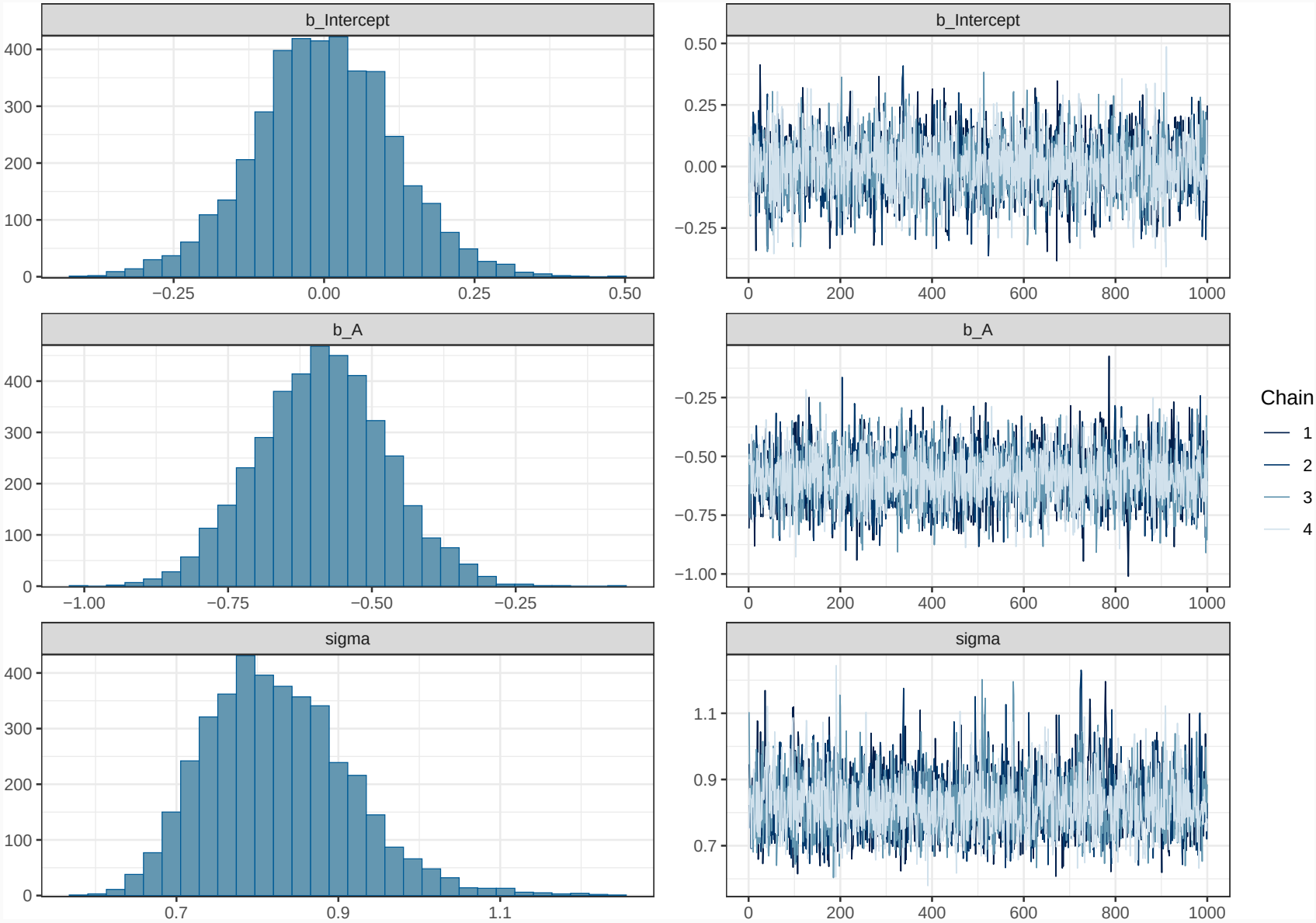
Evaluate the priors

```
# same priors on each model
p <- prior_draws(m_age)
ggplot(p,
       aes(x = 0, y = 0)) +
  geom_blank() +
  geom_abline(aes(intercept = Intercept, slope = b), alpha = 0.1) +
  coord_cartesian(xlim = c(-4, 4), ylim = c(-4, 4))
```



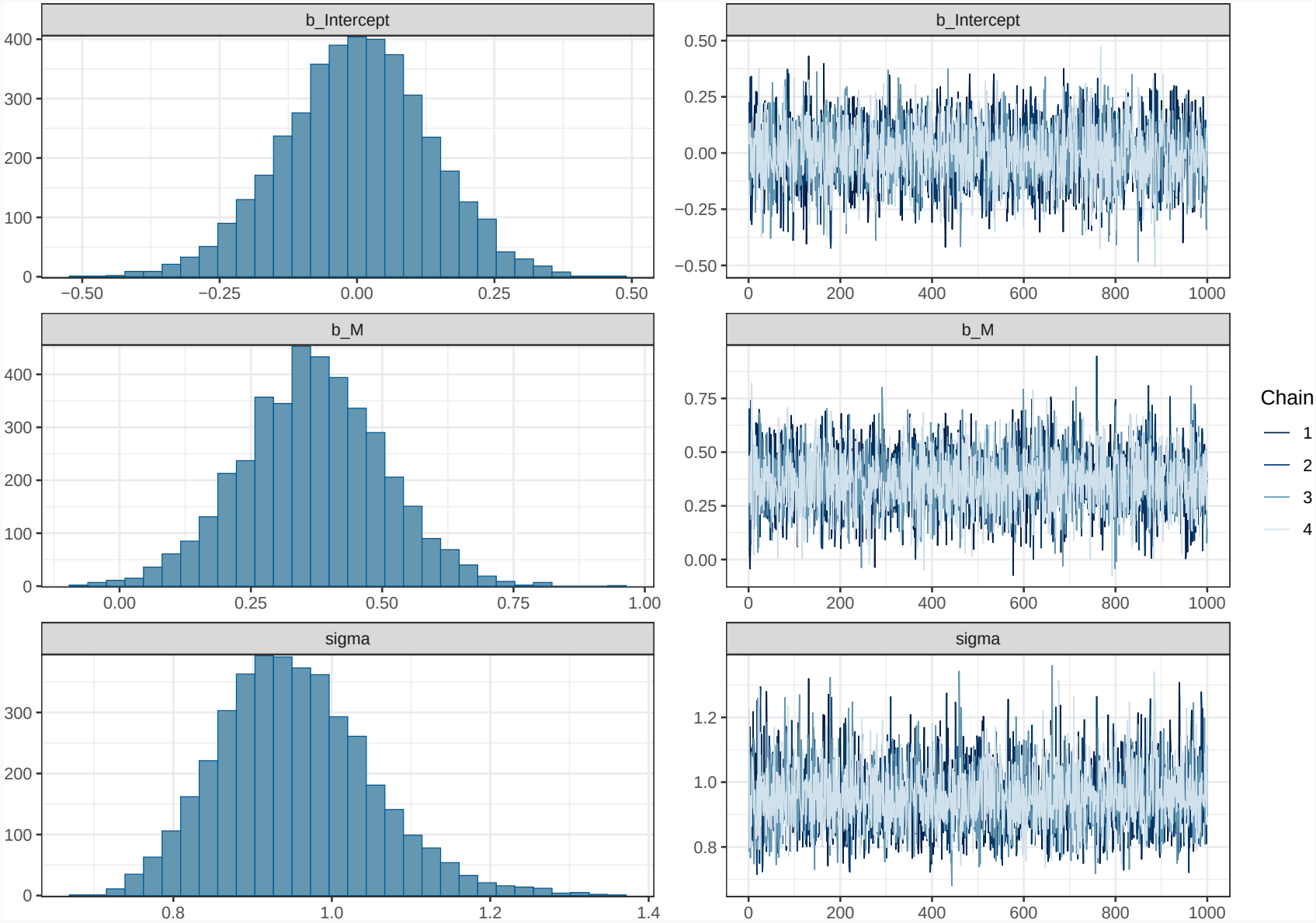
Evaluate the posteriors

```
plot(m_age)
```

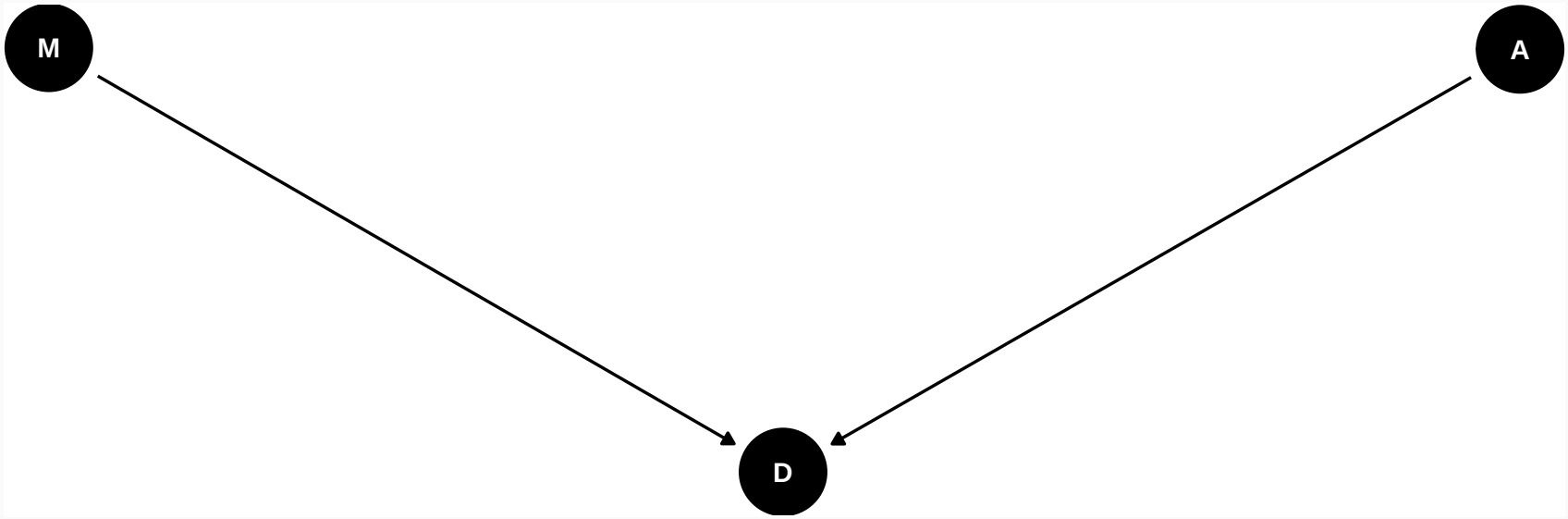


Evaluate the posteriors

```
plot(m_marriage)
```



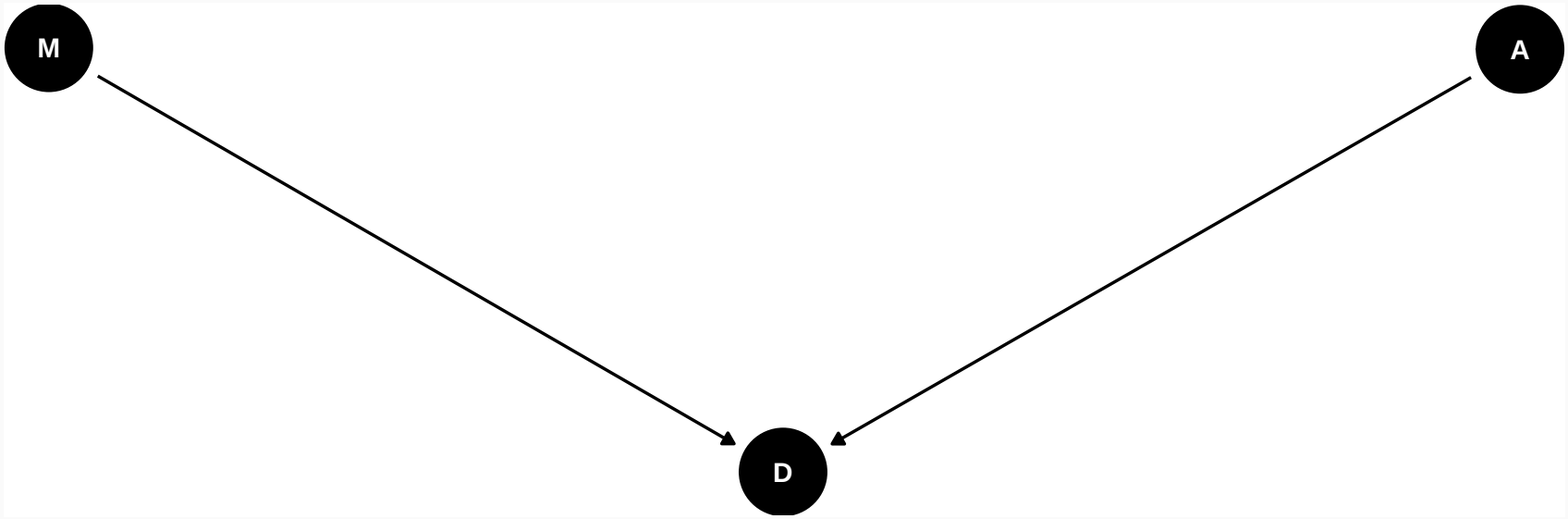
Testable implications of the DAG



`A _||_ M`

$$A \not\perp\!\!\!\perp D$$

Testable implications of the DAG

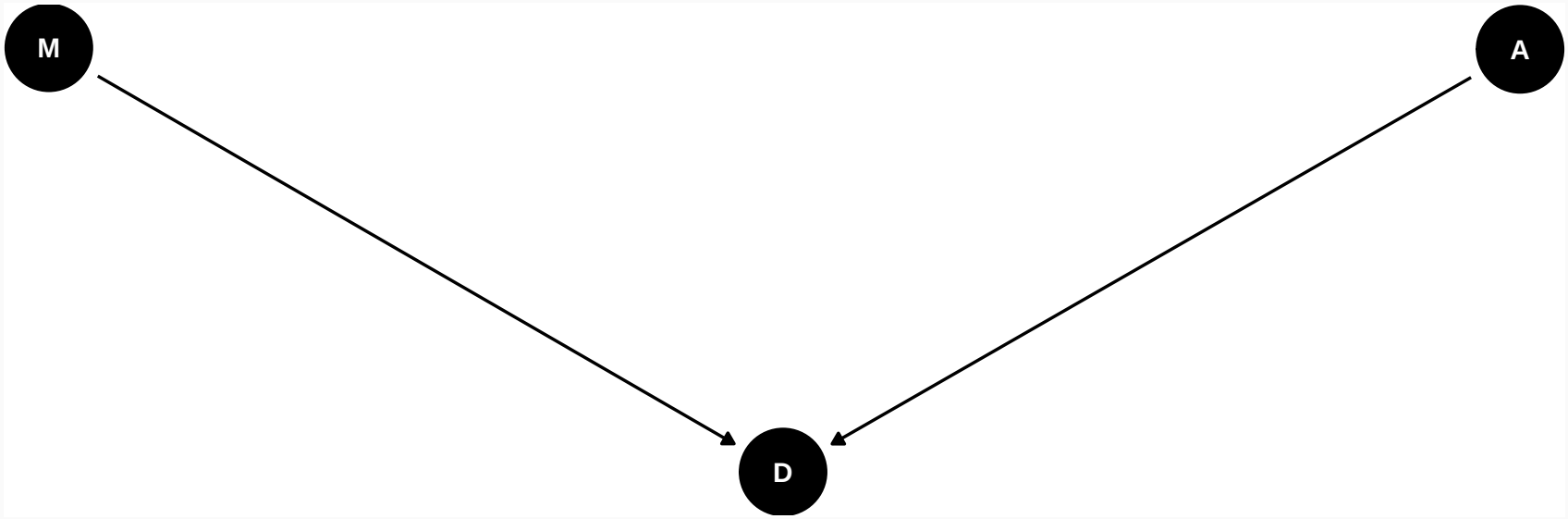



```
A _||_ M
```

$$A \not\perp\!\!\!\perp D$$

$$M \not\perp\!\!\!\perp D$$

Testable implications of the DAG



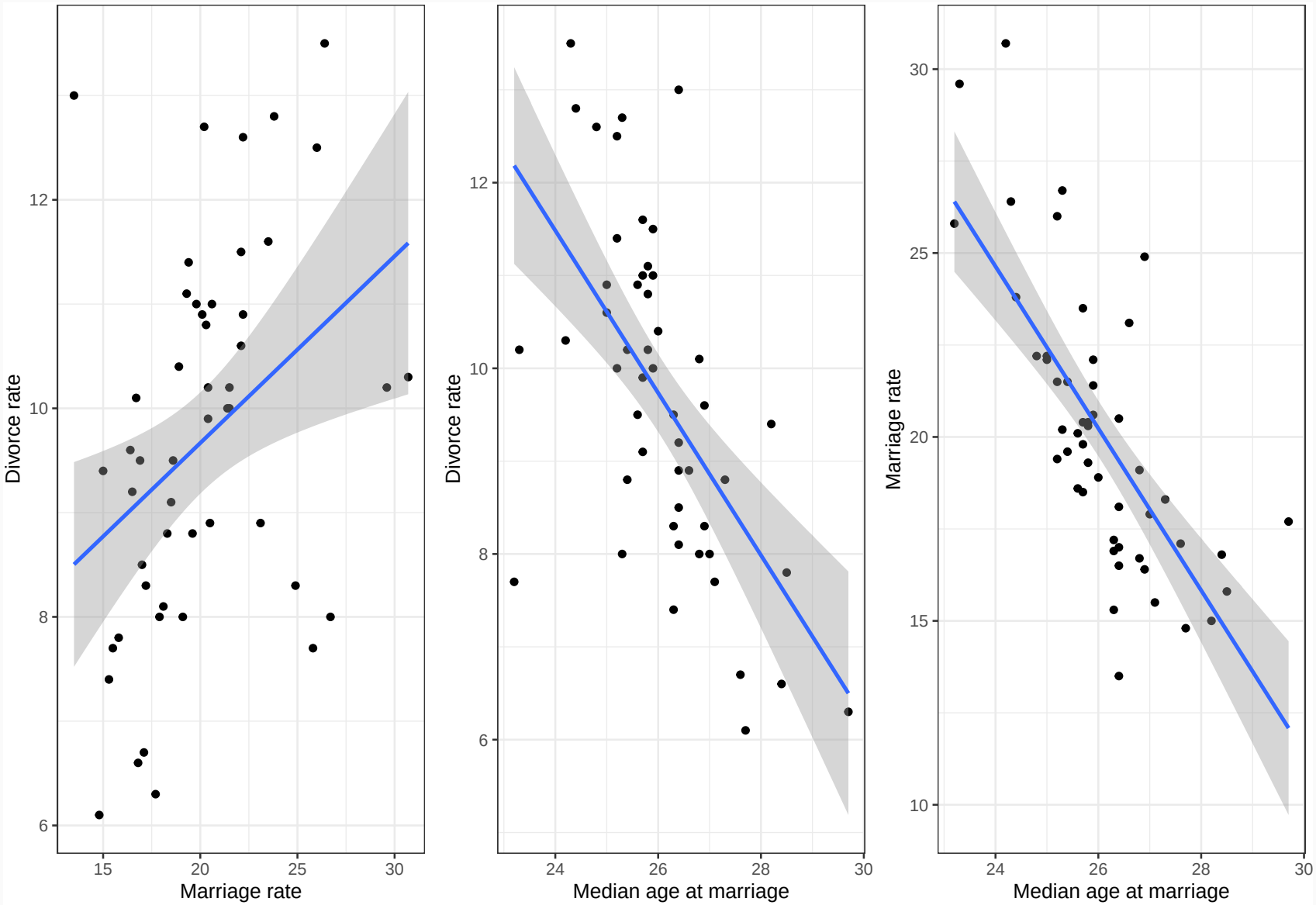

```
A _||_ M
```

$$A \not\perp D$$

$$M \not\perp D$$

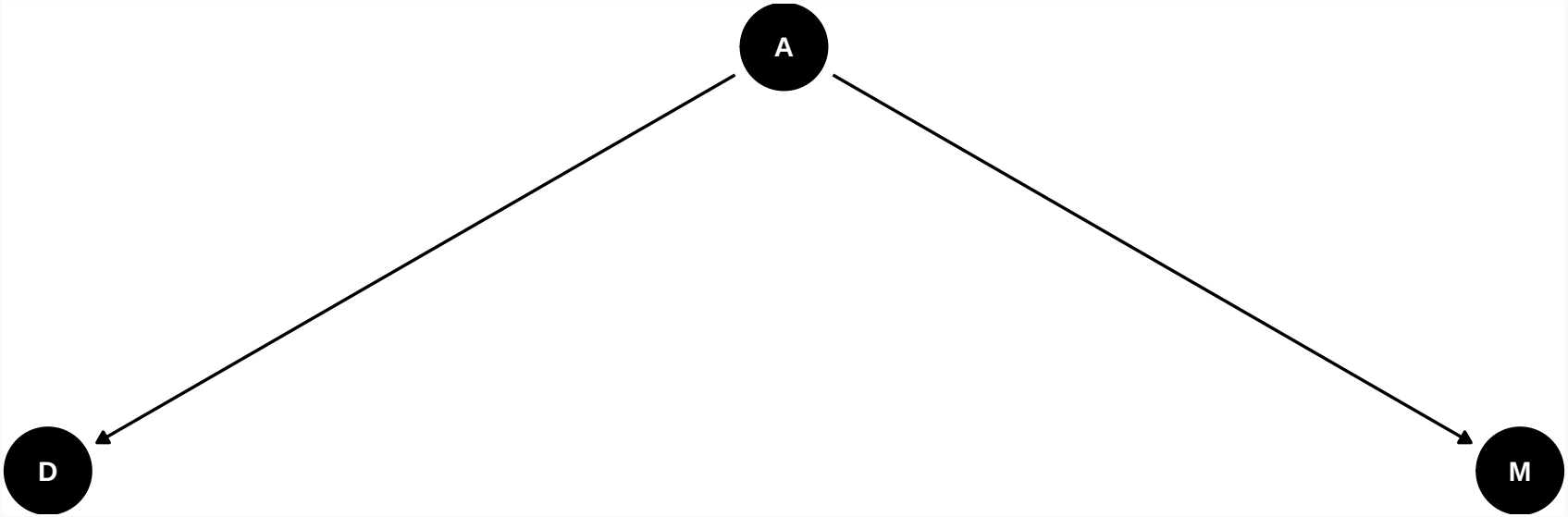
$$M \perp A$$

Graphical check



Revise our model

```
dagM <- dagify(D ~ A,
               M ~ A, outcome = "D")
ggdag(dagM) + theme_dag()
```




```
impliedConditionalIndependencies(dagM)
```

```
D _||_ M | A
```

This model says that divorce rates and marriage rates are independent of each other if we *stratify* by age at marriage.

Adding a second predictor

$$D_i \sim Normal(\mu, \sigma)$$

$$\mu_i = \alpha + \beta_M M_i + \beta_A A_i$$

$$\alpha \sim Normal(0, 1)$$

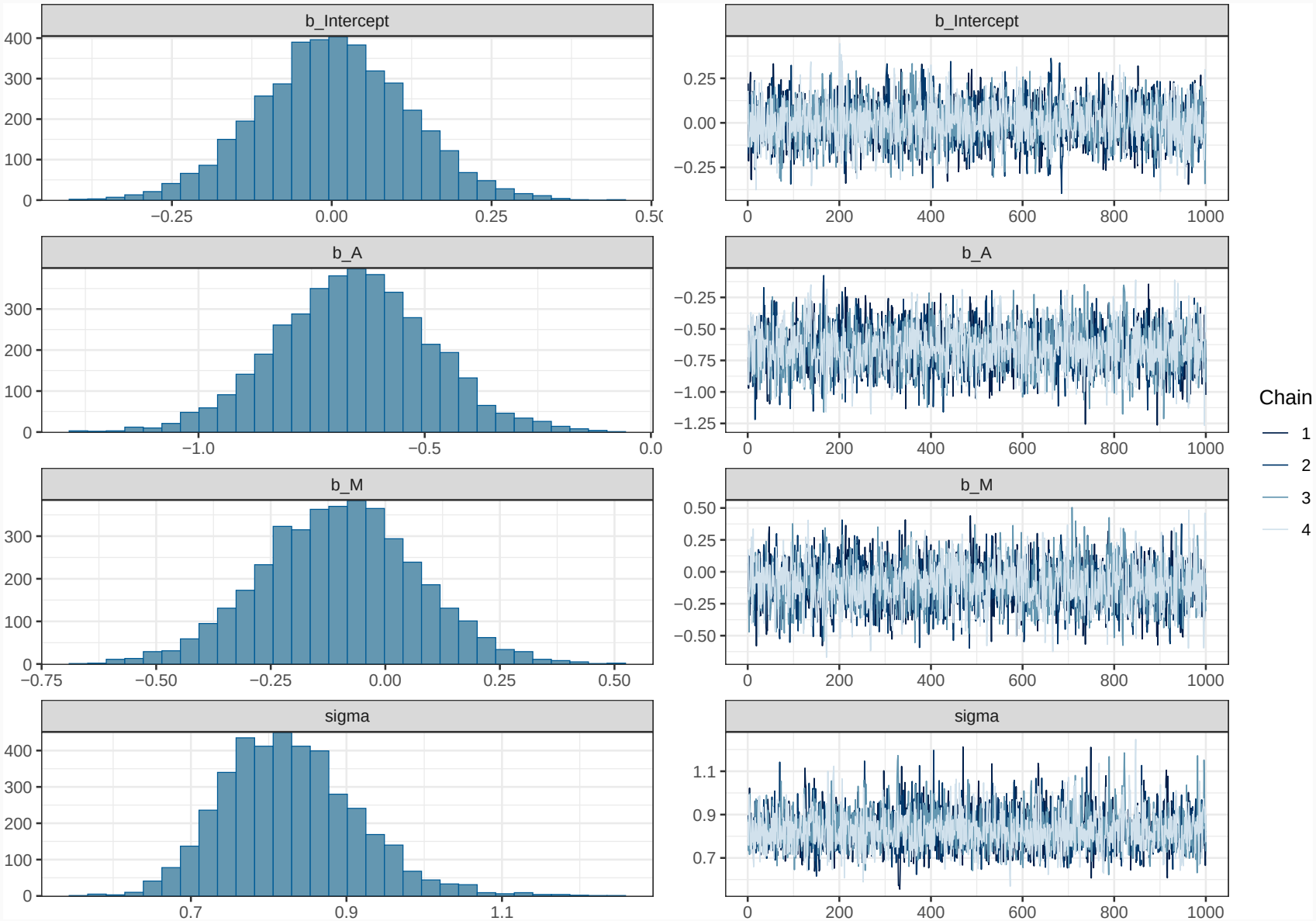
$$\beta \sim Normal(0, 1)$$

$$\sigma \sim Exponential(1)$$

Fit the model

```
m_both <- brm(D ~ A + M,
              prior = priors,
              data = WaffleDivorce)
```

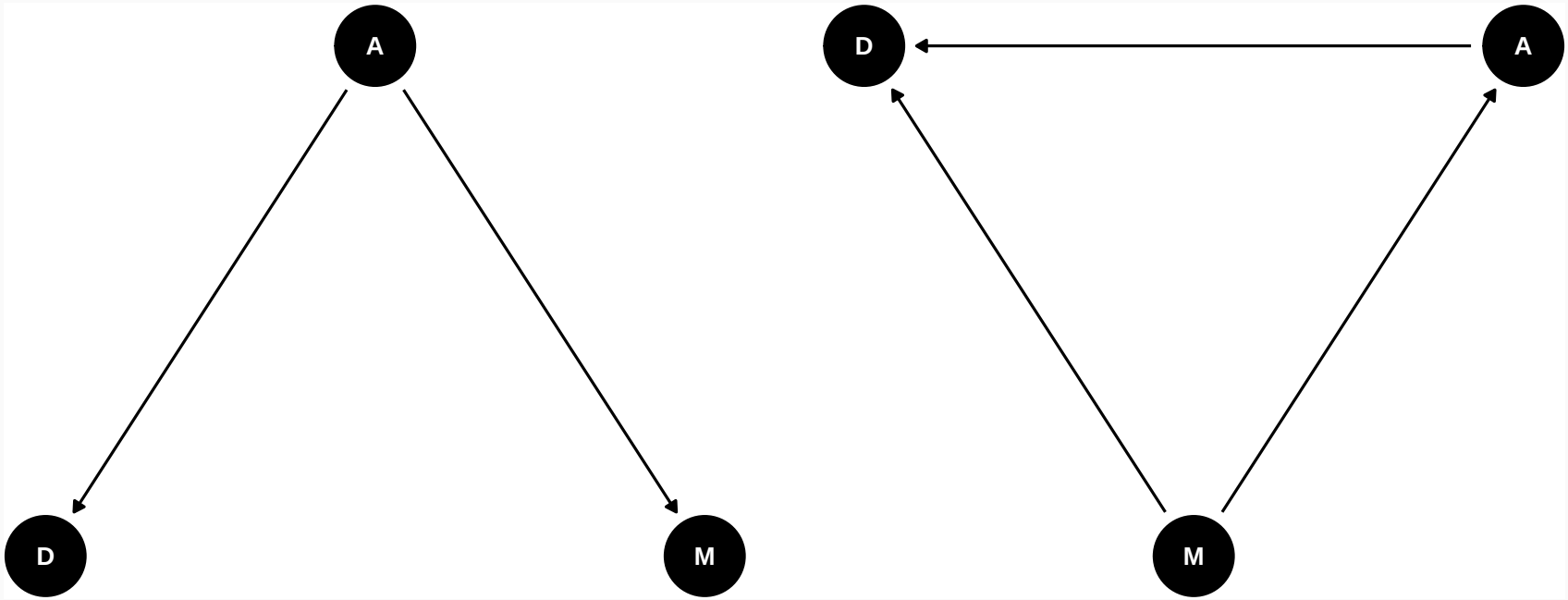
Visualize the posterior



Two DAGs are compatible with this regression

It is generally the case that many DAGs will be compatible with the same regression model

```
dag1 <- dagify(D~A,
               M~A, outcome = "D")
dag2 <- dagify(D~A+M,
                A~M, outcome = "D")
p1 <- ggdag(dag1) + theme_dag()
p2 <- ggdag(dag2) + theme_dag()
p1+p2
```



How to assess?

Look at model implications:

```
impliedConditionalIndependencies(dag1)
```

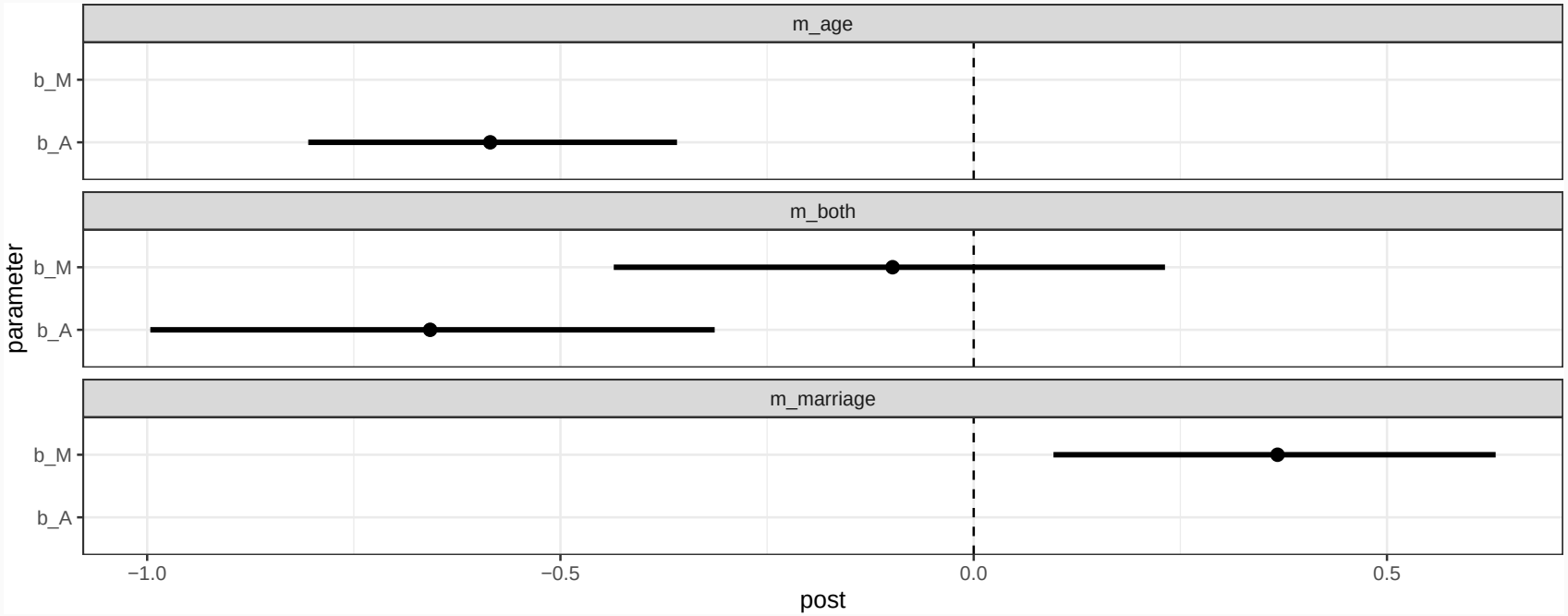
```
D _||_ M | A
```

```
impliedConditionalIndependencies(dag2)
```

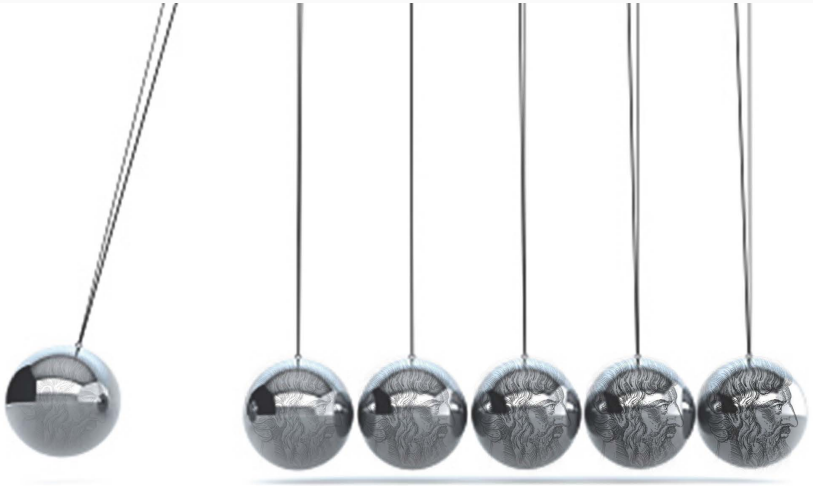
Model 1 says A is independent of M, conditional on A. Model 2 says all measures are associated.

Comparing results

Model 1 says A is independent of M, conditional on A. Model 2 says all measures are associated.



Recommended reading



CAUSAL INFERENCE IN STATISTICS

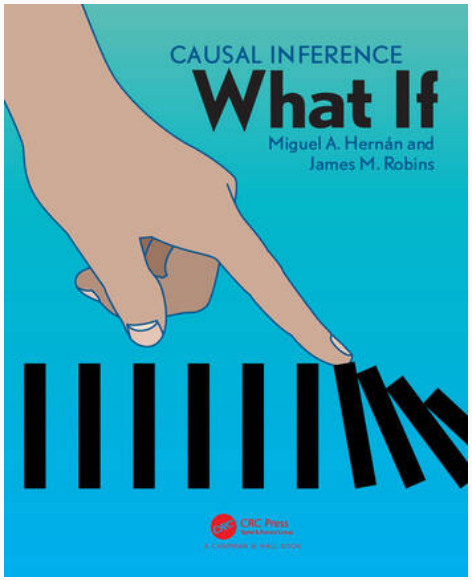
A Primer

Judea Pearl
Madelyn Glymour
Nicholas P. Jewell

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Recommended reading



And accompanying free EdX course (it's good!)

https://harvardonline.harvard.edu/course/causal-diagrams-draw-your-assumptions-your-conclusions